autoneum

— Confidential —

Application of release agents at the Ellzee plant

Development of new technology for CO2-based RA

E.Jeckle

11.06.2024

autoneum

— Confidential —

Agenda

1. Different ways of demoulding
2. Application and function of RA
3. Layer structure of RA in the mold
4. Presentation CO2-Technology - Ingredients
5. Presentation CO2-Technology - Flowchart
6. Presentation CO2-Technology - Pictures of special technology and equipment
7. Immediate effects and future savings potential
8. Cost development at the Ellzee plant 2015-2022 during the conversion from aqueous to CO2-based RA
9. Consequences of not pursuing CO2 technology
10. Comparison of running costs Comparison of CO2 against aqueous

1. Different ways of demoulding

autoneum

There are different types of release agents

Demoulding with foil	High waste and necessary of vacuum technology
Demoulding with plastic cavities	Additional effort and costs for injection moulded insterts
Permanent coating of the tool	No permanently functioning and practical/economic technology

Using release Agents	
Solvent based RA´s	High emissions
Water based RA´s	Industry standard with high consumption and cleaning efforts
CO2 based RA´s	low consumtion, cleanig efforts, emissions, high invest. **Ellzee Standard**

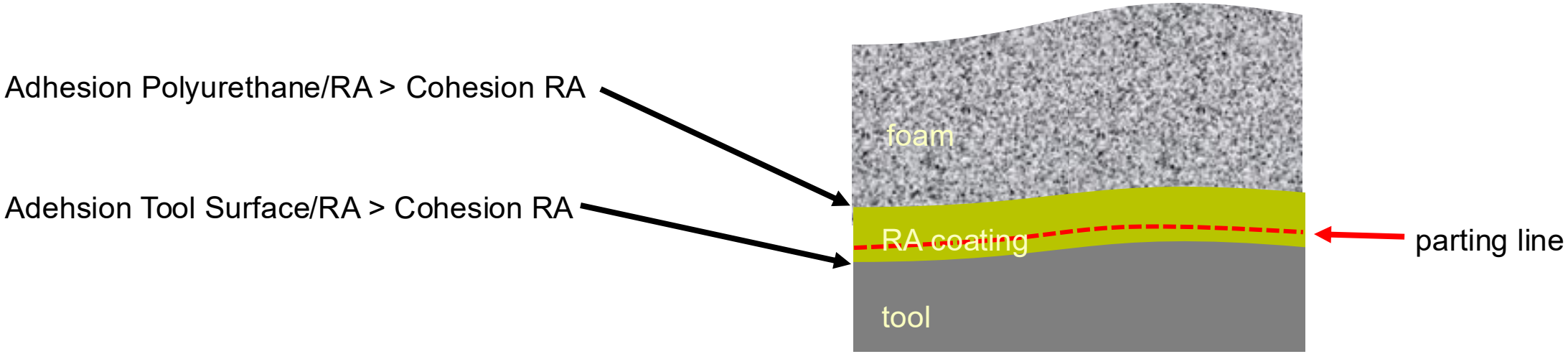
2. Release agent underestimated support material

Application and Function

Many industrial productions and applications are impossible without release agents. This is also the case with many of our processes in the automotive sector. In the following, we deal with release agents for polyurethane foams, especially in sensitive flexible foam.

As a coating, the release agent layer, is introduced into the tool.

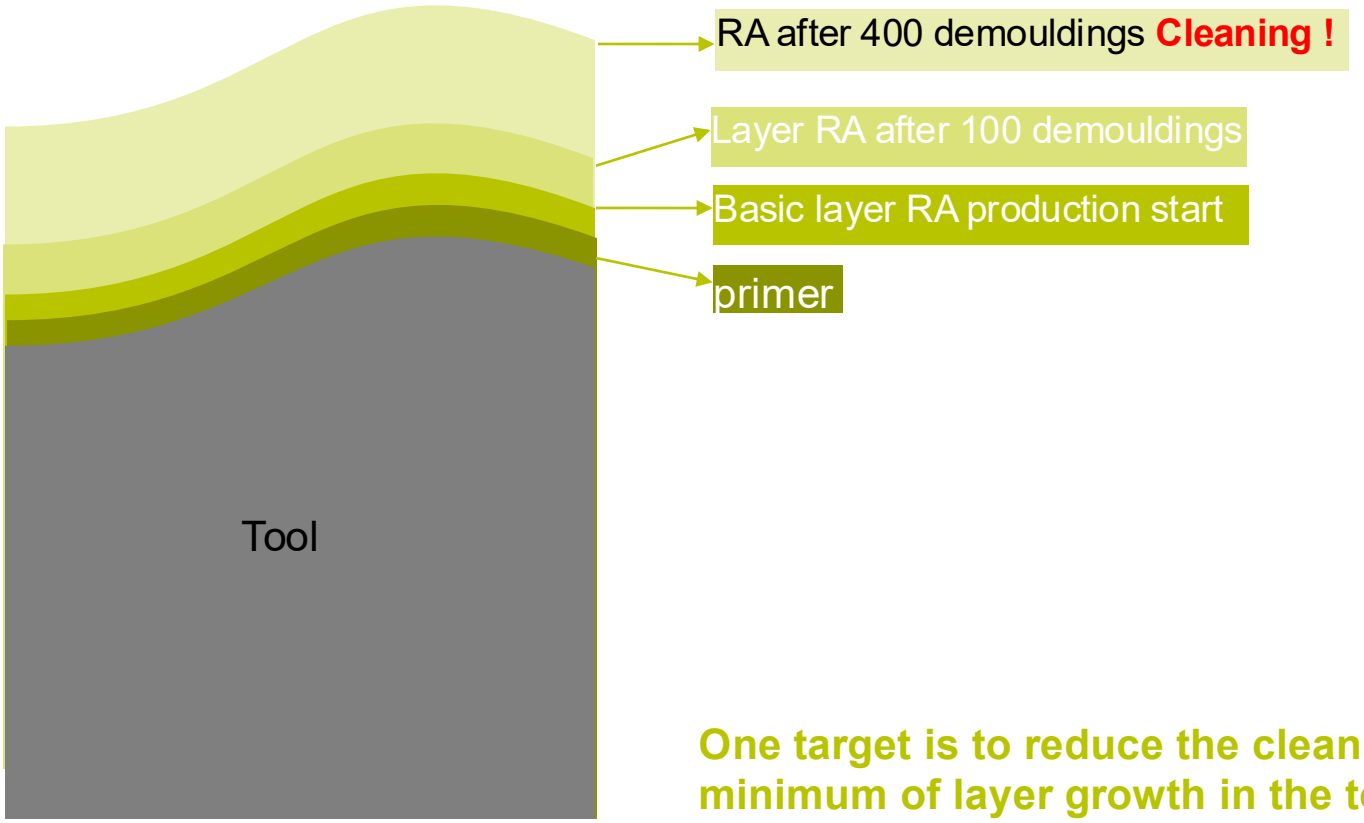
This layer makes component production in a mold possible in the first place. It is important to note:



These conditions place the parting line in the middle of the release agent layer during the demolding process. This means that half of the RA remains on the component side, the other half remains in the tool. When one of this conditions will not function, the demoulding would run in complete different way, till to destroying the demoulded part.

3. Release agent build-up in the mold

Layer structure up to the cleaning cycle



After we had cleaned the tool, normally by dry ice we have to prepeare the tool surface, before we start the foaming process again.

For this reason we first apply a primer, a release concentrate. We do this via normal spraying process without CO2.

Then we apply a few layers of our standard release agent.

Now we start producing.

While the prduction process the layer of the released agent is growing.

After a lot of of demouldings we have to clean the tool again.

One target is to reduce the cleaning cycles to a minimum, this means only a minimum of layer growth in the tool.

4. Presentation of CO2-based RA technology

autoneum

Composition RA

An RA is a composit up of several components which have necessarily be adapted to the ingredients of the foam to be separated:

- separating substances such as paraffins and oils,
- Water
- other solvents based on aromatic and aliphatic compounds, which represent emissions in the sense of VOC.
- Additives for stabilizing, cell opening, surface curing.

The water has the function of a support matrix, carrying the other substances. Hence the name aqueous RA.

In the CO2-based technology, the carrier water is exchanged for liquid CO2. The VOC-laden solvents can be significantly reduced once again.

The other additives are adapted accordingly. The entire composite is much more concentrated than in aqueous RA.

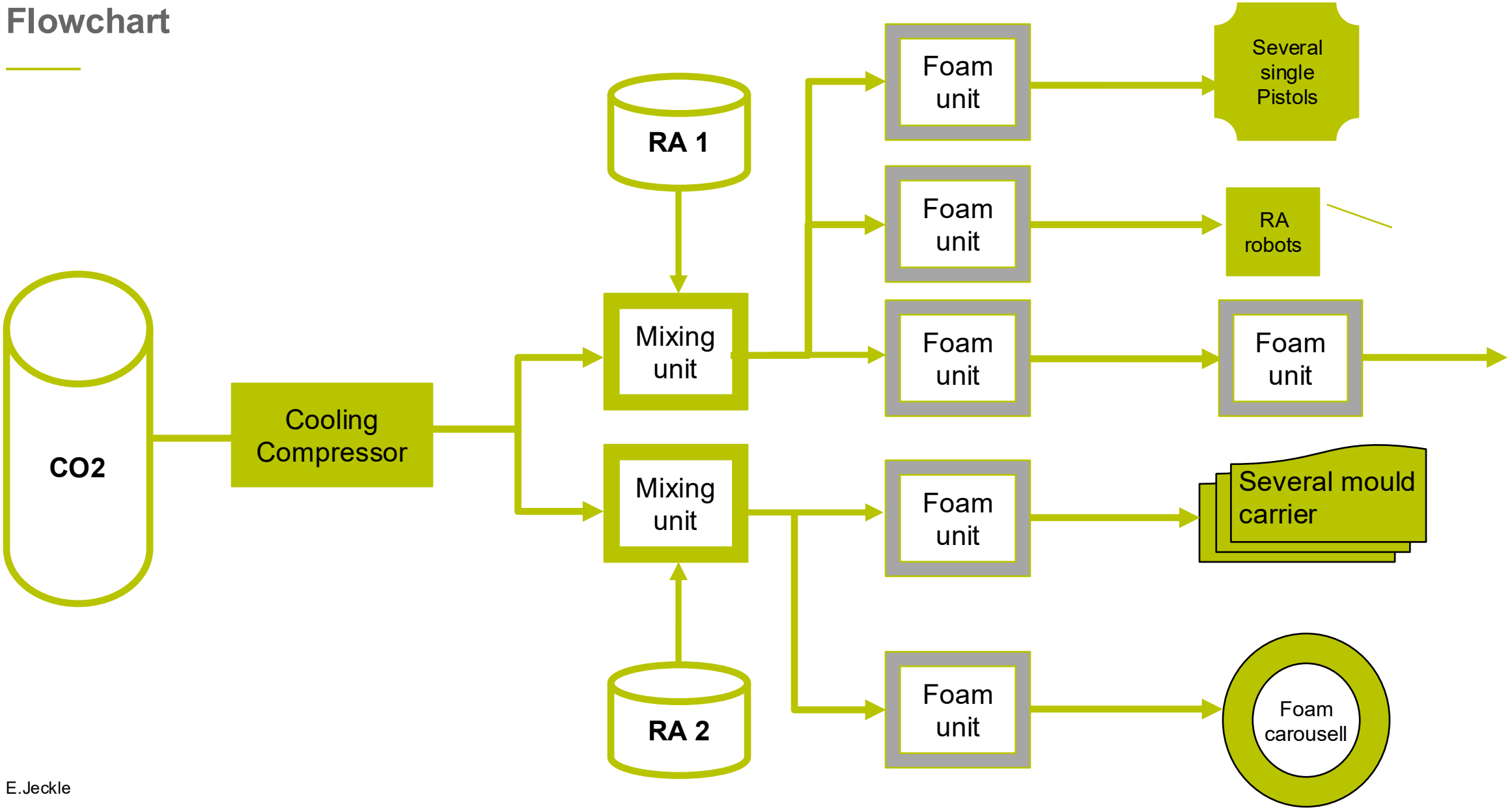
The RA is used in diluted form by diluting it with liquid CO2 via a mixing system and transporting it to the production units via high-pressure pipes.

Using standard spray guns, the CO2 is emitted as a white mist, while the RA is thin coating the mould surface.

5. Presentation of CO2-based RA technology

autoneum

Flowchart

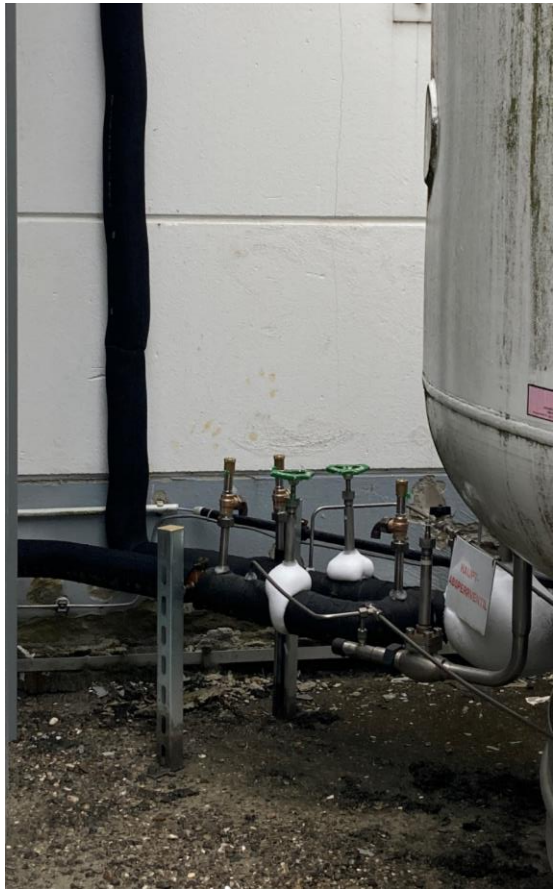


6. Presentation of CO2-based RA technology

autoneum

Pictures of the special Equipment

Liquid-Gas Tank for CO2



Distributor-fitting for CO2, thermally insulated

High pressure lines and Distributor valves, gas taps inside the Manufactoring halls.

Compressor deep-freeze unit, mixing unit

7. Project Forecast

Development of a mixer standard for CO2-based RA

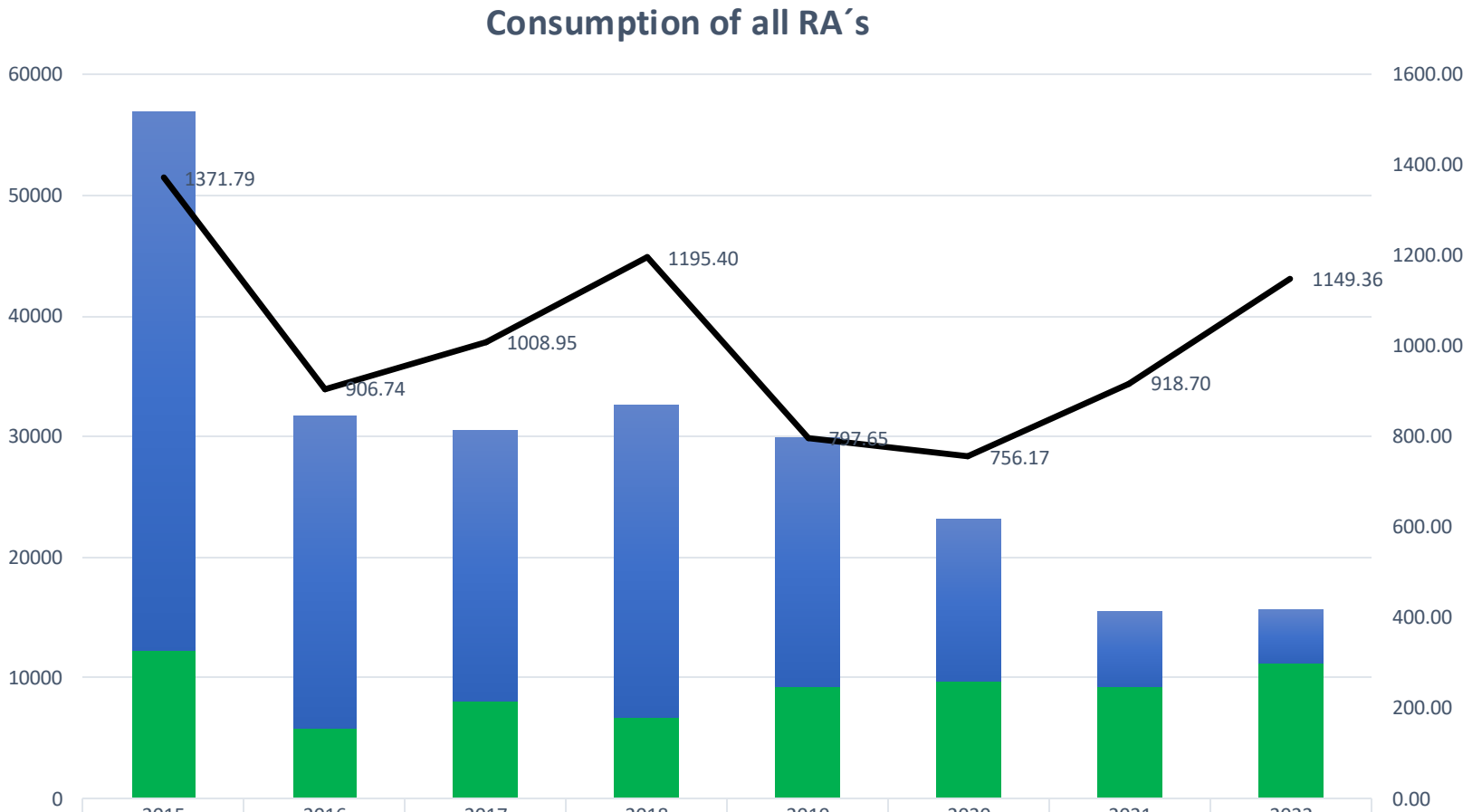
autoneum
— Confidential —

Savings Forecast, Consumption/Costs and Secondary Effects
Immediate Effects
Use of state-of-the-art spraying technology leads to a reduction in consumption and time in the start-up or priming phase.
Reduced RA consumption of up to 60-70% due to innovative technology (later up to 80% next gen. of mixing units)
Significantly extended cleaning cycles, i.e. less cleaning by 0.5-1.5 cycle/week
Cycle time shortening due to reduced venting time.
Reduced scrap by too much using RA.
Further potentials
Further savings potential through the use of new Release Agents which are even further reduced by 10-20% (not VOC-relevant)
Transfer of knowledge from flexible foam to rigid foam. Potential cost savings of up to 50%
Transfer of the technology to other Autoneum plants.

8. Overview cost development in Ellzee plant 2015 -2022

Switching from aqueous RA to CO2-based RA

autoneum
— Confidential —



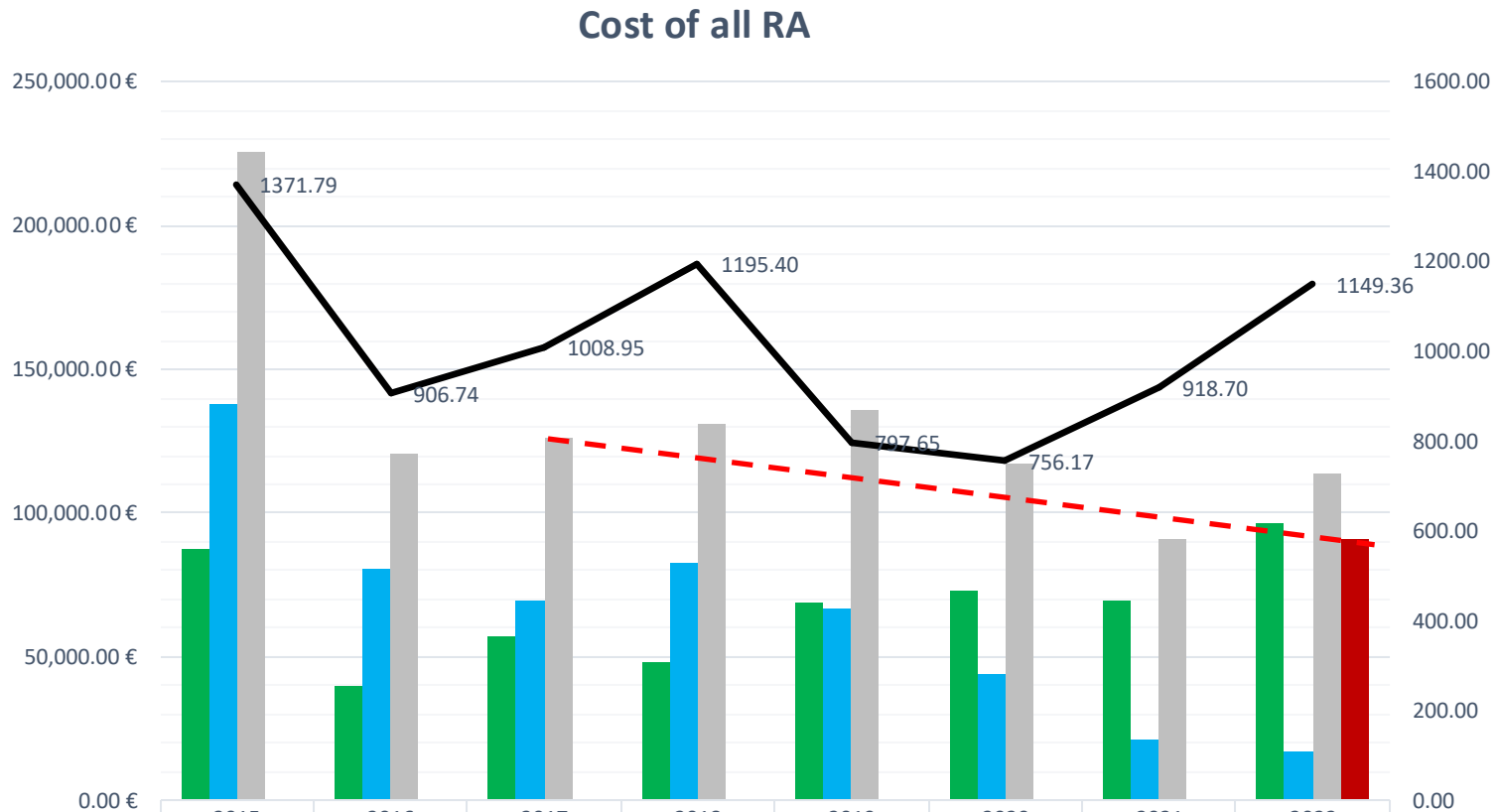
	2015	2016	2017	2018	2019	2020	2021	2022
kg aqueous	44711	26100	22500	26100	20647	13500	6300	4420
kg auf CO2	12300	5700	8100	6600	9260	9740	9293	11250
foam in to	1371.79	906.74	1008.95	1195.40	797.65	756.17	918.70	1149.36

The figures are overlaid by economic developments, order start-ups and outflows.

8. Overview cost development in Ellzee plant 2015 -2022

Switching from aqueous RA to CO2-based RA

autoneum
— Confidential —



	2015	2016	2017	2018	2019	2020	2021	2022
€ CO2	87,645.00 €	39,855.00 €	57,015.00 €	47,987.82 €	69,056.20 €	73,133.80 €	69,258.69 €	96,727.50 €
€ aqueous	138,156.99 €	80,649.00 €	69,525.00 €	82,737.00 €	67,102.75 €	43,875.00 €	21,464.73 €	16,796.00 €
€ total	225,801.99 €	120,504.00 €	126,540.00 €	130,724.82 €	136,158.95 €	117,008.80 €	90,723.42 €	113,523.50 €
prize 2017								90,818.80 €
foam production	1371.79	906.74	1008.95	1195.40	797.65	756.17	918.70	1149.36

Until 2017, the cost situation was absolutely stable. From 2017 onwards, consistent price increases were made.

2017-2022 cumulative, the increase is 20-22%

Adjusted for the price increases, a saving of 20-30% was realized.

This would correspond to approx. 35,000 €

The figures are overlaid by economic developments, order start-ups and outflows.

9. Consequences of not pursuing future production with CO2

Provision to aqueous-based release agent

autoneum
— Confidential —

Consequences by switch to water based system

- Gradual relapse to water-based RA.
  The existing CO2 units would no longer be replaced or repaired.
  The former aqueous RA is no longer produced, so an immediate search for suitable RA with production trials would have to be initiated. Conversion period across all systems: at least 1 year

- Risk of increased solvent/VOC emissions of up to 3 tons/year, as temporarily hybrid RA cannot be ruled out.

- Significantly increased RA consumption, 3-4 times (experience from the past + confirmation from Chemtrend)

- Increased number of rejects and reduced qualities for complex components,

- cycle time extensions, as much more liquid has to air off in the cycle.

- Significantly increased tool cleaning of up to 50-100% (it depends on turn over and size of the tool)
  The increased RA build-up in the Wkzg. shortens the cleaning cycles by up to half. The consistency of the "overspray" leads to rock-hard encrustations on the outside of the plant, mould carrier, and periferia which must be removed with additional effort and material wear.

- Additional downtimes (cleaning times + additional downtimes) Start-up times with primer (45-55 min./wkzg and cycle) reduce machine capacities.

10. Comparison of CO2 base and return on aqueous

Comparison running costs for the Ellzee plant

autoneum
— Confidential —

Forecast of costs provision on an aqueous basis	
TM annual. Costs of flexible foam on a cost basis Q2/2023 approx. 97.000€	
TM annual. Costs of flexible foam after provision from aqueous approx. 137.000€	(36 to with 3.8€)
Additional annual Cleaning costs flexible foam 50-100% 40000€ - 60000€	(today approx. 10000€/month)
Additional costs RA flexible foam / year : **75000,-€ - 95000,-€**	
Secondary costs (cycle extension, cappared., rejects, etc.) + X ,-€	

Autoneum. Mastering sound and heat.

Erich Jeckle

PU-Schaumentwicklung Autoneum Werke Krumbach-Ellzee

Erich.jeckle@autoneum.com
+498282/93371